INCH-POUND

MIL-C-55514/11A(ER)
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SUPERSEDING
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MILITARY SPECIFICATION SHEET

CAPACITORS, FIXED, METALLIZED PLASTIC DIELECTRIC, DC, IN NONMETAL CASES, ESTABLISHED RELIABILITY STYLES CFR26 AND CFR27

This specification is approved for use by the Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-55514

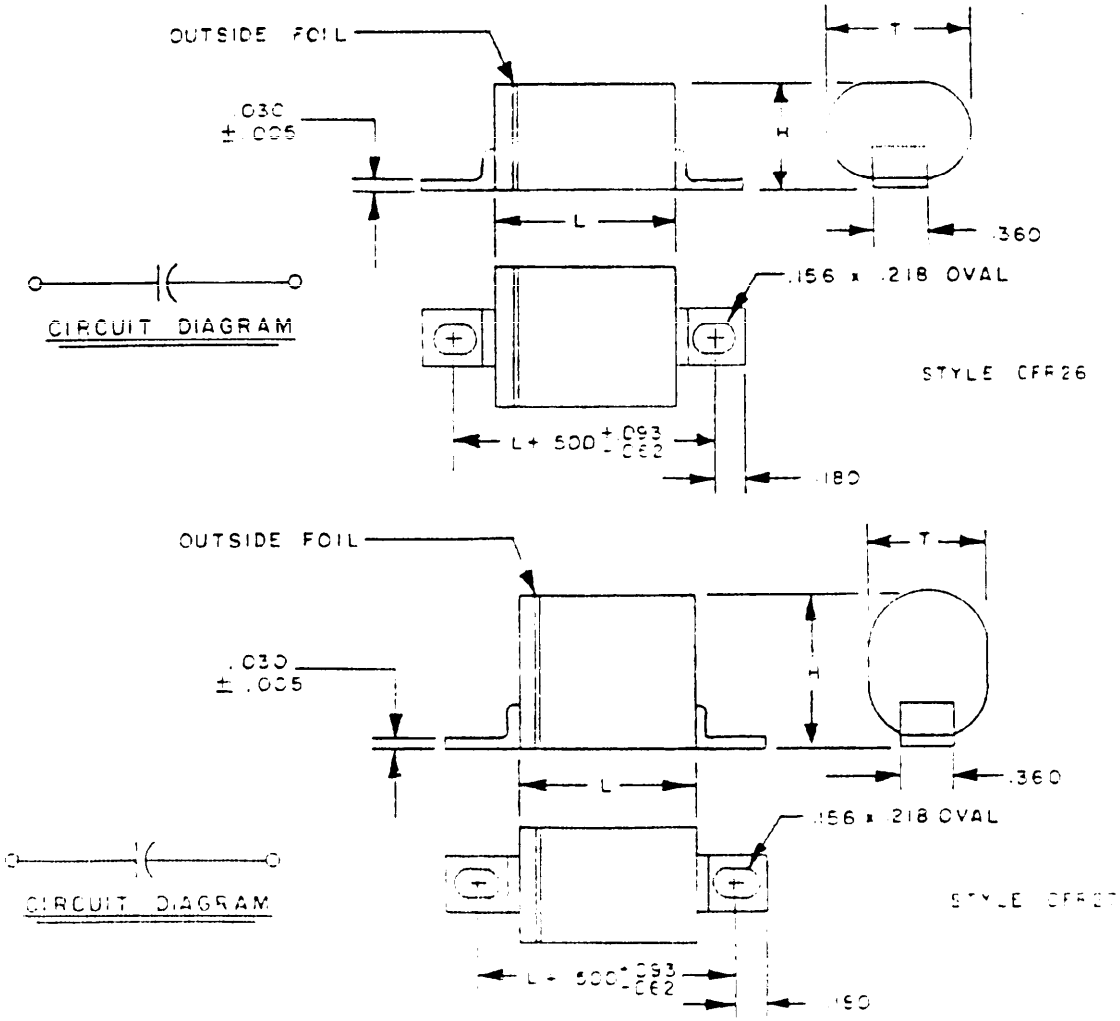


FIGURE 1. Styles CFR26 and CFR27 capacitors.

AMSC N/A FSC 5910

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Inches	mm
.005	0.13
.030	0.76
.046	1.17
.062	1.57
.093	2.36
.156	3.96
.180	4.57
.218	5.54
.360	9.14
.437	11.10
.468	11.89
.500	12.70
.562	14.27
.593	15.06
.718	18.24
.750	19.05
.781	19.84
.812	20.62
.875	22.23
.906	23.01
.937	23.80
.968	24.59
1.000	25.40
1.093	27.76
1.250	31.75
1.281	32.54
1.312	33.32
1.437	36.50
1.687	42.85
1.812	46.02

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.010 (0.25 mm).
4. These capacitors are to be mounted by screws.

FIGURE 1. Styles CFR26 and CFR27 capacitors - Continued.

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REQUIREMENTS:

Design and construction:

Dimensions and configuration: See figure 1.

Case: Nonhermetic.

Dielectric: Metallized polycarbonate.

Characteristic: R.

Capacitance value: See table I.

Capacitance tolerance: See table I.

Solderability: Not applicable.

Operating temperature range: -55°C to +85°C, when properly voltage derated (see figure 2), characteristic "R" capacitors will operate to +125°C.

Voltage rating: See table I.

Preconditioning: Not applicable.

Burn-in: In accordance with MIL-C-55514 except for the following conditions:

a. DC burn-in voltage shall be 140 percent of the dc rated voltage applied for 48 hours minimum at 125°C +4°C, -0°C.

b. AC burn-in: Not applicable.

Insulation resistance: Insulation resistance shall be measured at rated voltage at the applicable temperature. Charge time 2 minutes maximum; however, for capacitance values greater than 1.0 µF, an additional 1 minute per µF is permitted. The following limits shall apply:

Temperature	Insulation resistance
25°C	50,000
85°C	5,000
125°C	500

Marking: In accordance with MIL-C-55514.

Qualification submittal:

Type designation	Sample size
CFR27LRA106JM	61
CFR27LRA506JM	61
CFR27LRG505JM	61
CFR27LRG506JM	61
CFR27LRB505JM	61
CFR27LRB506JM	61

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TABLE I. Capacitor characteristics and dimensions.

Type designation 1/	Capacitance	Capacitance tolerance	Rated voltage at 85°C	ESR milliohms at 100 kHz	Ripple current max at 25°C	Resonant frequency kHz	Style CFR26			Style CFR27		
							T 2/	H 2/	L 3/	T 2/	H 2/	L 3/
CFR2-LRA106--	10.0 µF	J K M	50 V	11	12.6	265	.812	.562	.750	.562	.812	.750
CFR2-LRA206--	20.0 µF	" " "	50 V	10	14.8	178	.968	.718	.875	.718	.968	.875
CFR2-LRA506--	50.0 µF	" " "	50 V	6	23.5	108	1.375	1.000	1.000	1.000	1.375	1.000
CFR2-LRG505--	5.0 µF	J K M	75 V	13	11.6	356	---	---	---	.500	.750	.875
CFR2-LRG106--	10.0 µF	" " "	75 V	12	13.5	251	1.031	.718	.875	.781	1.031	.875
CFR2-LRG206--	20.0 µF	" " "	75 V	11	16.5	171	1.312	.937	1.000	.937	1.312	1.000
CFR2-LRG506--	50.0 µF	" " "	75 V	6	24.2	100	1.687	1.312	1.250	1.312	1.687	1.250
CFR2-LRB505--	5.0 µF	J K M	100 V	13	12.4	356	.906	.593	.875	.593	.906	.875
CFR2-LRB106--	10.0 µF	" " "	100 V	12	14.6	242	1.093	.781	1.000	.781	1.093	1.000
CFR2-LRB206--	20.0 µF	" " "	100 V	11	17.1	160	1.281	.906	1.250	.906	1.281	1.250
CFR2-LRB506--	50.0 µF	" " "	100 V	6	23.6	96	1.812	1.437	1.375	1.437	1.812	1.375

1/ The complete type designation will include additional symbols to indicate style, capacitance tolerance, and failure rate level (M, P, R, and S).

2/ Tolerances: Thickness and height +.062, -.156 on .500 and +.093, -.187 on .562 and greater.

3/ Tolerances: Length +.093, -.062.

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Substitutability for failure rate levels shall be as follows:

Failure rate level	Will replace failure rate level
S	R, P, M
R	P, M
P	M
M	

Substitutability for capacitance tolerances shall be as follows:

Capacitance tolerance	Will replace capacitance tolerance
J	K, M
K	M
M	

Substitutability for voltages shall be as follows:

Volts dc	Will replace voltage
100	75, 50
75	50

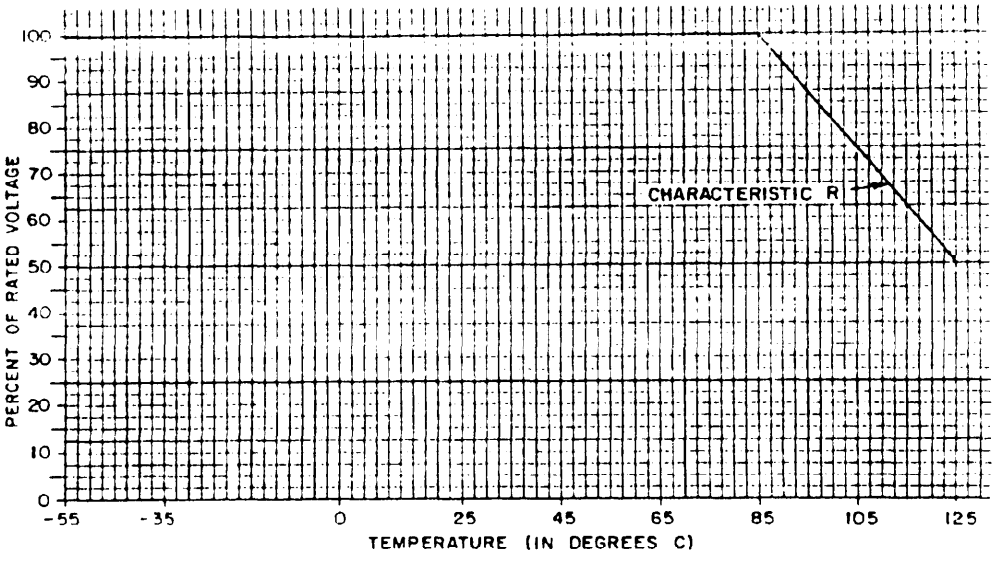


FIGURE 2. Voltage derating with temperature.

NOTE: Revision letters are not used to denote changes due to the extensiveness of the changes.

CONCLUDING MATERIAL

Preparing activity:
Army - ER

Agent:
DLA - ES

(Project 5910-A440)

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